Forensic Building Envelope Analysis: Ventilation Failure & Thermal Tracking

Date: January 10, 2026
Roof install date: April 12, 2024

Subject: Roof Ventilation Imbalance & Interior Environmental Failure

Property Type: 1,700 sq. ft. Rancher (4/12 Slope), Victoria, BC (Zone 4C)

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1. Executive Summary

A forensic review of the recently installed roof system identifies a critical failure in the ventilation strategy. The installation of high-volume continuous exhaust (Omni Ridge Vent) paired with restricted, non-compliant intake (Menzies SV-50) has created a state of **Attic Depressurization (Negative Pressure Gradient)**.

This negative pressure is actively pulling conditioned indoor air through the ceiling plane, resulting in "Thermal Tracking" (ghosting) along joist lines. The intake vents currently installed are **prohibited by the manufacturer** for this specific application, constituting a breach of installation standards.

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2. Quantitative Analysis: The Flow Imbalance

While the *total* amount of ventilation may technically meet the minimum square footage requirement of IRC R806.2, the system violates the **physics of balanced ventilation**. Best practices dictate a 50/50 balance or a 60/40 intake-heavy ratio.

A. Exhaust Capacity (The Drive)

- **Product:** Omni Ridge Vent (Continuous)
- **Spec:** 18 sq. in. Net Free Area (NFA) per linear foot.
- **Quantity:** 60 linear feet.
- **Total Exhaust NFA:** [1]$60 \times 18 =$ **1,080 sq. in.**[2]

B. Intake Capacity (The Choke)

- **Product:** Menzies Metal Products SV-50 Plastic Roof Vent.
- **Spec:** 50 sq. in. NFA per unit (Nominal).
- **Quantity:** 12 units.
- **Theoretical Intake NFA:** $12 \times 50 =$ **600 sq. in.**

- **Real-World Effective NFA:** Due to the 4/12 slope and "reverse flow" turbulence (using an exhaust vent for intake), effective capacity is estimated to be reduced by **40-60%**.
  - *Estimated Effective Intake:* **~300 - 360 sq. in.**

C. The Deficit

- **The Ratio:** The Exhaust capacity exceeds Intake capacity by a factor of roughly **3:1**.
- **Status: Severe Depressurization.** The ridge vent acts as a vacuum pump. To satisfy the airflow demand, it pulls the missing air volume from the house interior, bypassing the restricted roof vents.

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3. Product Compliance & Legal Standard

The primary failure point is the misuse of the **Menzies SV-50** vent.

- Manufacturer Prohibition:

  The Menzies SV-50 Product Manual explicitly states: "Not to be used as intake vent." The internal baffles are designed to shed rain while exhausting air; forcing air inwards creates turbulence and water ingress risks.

- Slope Violation:

  The product is designated for "Steep Roof Application" (Minimum 3.5/12 or 4/12 depending on revision). Using a static vent on the eave of a low-slope roof places the vent in the wind boundary layer, where it often experiences neutral pressure, rendering it ineffective as an intake.

Legal Standard: The use of products contrary to manufacturer instructions constitutes a definitive installation failure. Manufacturer prohibitions define the scope of warranty coverage and professional liability; they cannot be overridden by "common practice."

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4. The Failure Mechanism: "Thermal Tracking"

The sudden appearance of black lines (Ghosting) immediately following the roof installation provides the temporal evidence of failure.

1. **Pressure Differential:** The unbalanced Ridge Vent creates a negative pressure (ΔP) in the attic relative to the living space.
2. **Infiltration:** This suction pulls warm, particulate-laden indoor air through the ceiling insulation (via pot lights, wire penetrations, and top plates).

3. **Deposition:** As this dirty air passes over the cold ceiling joists (Thermal Bridges), the particles (soot/dust) condense and stick to the drywall.[3]
4. **Causality:** The roof change was the only variable. The home performed for 20 years without ghosting because the previous roof did not induce high-suction negative pressure.

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5. Anticipated Defenses & Rebuttals

- **Defense:** *"This is common practice; we use these all the time on low slopes."*
  - **Rebuttal:** "Common practice" does not override the **Manufacturer's Written Prohibition**. Additionally, the immediate onset of ghosting proves that while it may work on leakier houses, it has failed on this specific building envelope.
- **Defense:** *"The ghosting is caused by candles or a dirty house."*
  - **Rebuttal:** The ghosting appeared **immediately** after the roof installation. Occupant habits (cooking/cleaning) did not change overnight. The only changed variable is the roof pressure system.
- **Defense:** *"The total NFA meets code."*
  - **Rebuttal:** Meeting the *total* NFA minimum does not excuse the *imbalance* or the *misuse of product*. A system that meets code but destroys the interior via depressurization is still a failed system.

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6. Remediation Requirements

To stop the interior damage and restore building envelope integrity, the following is required:

1. **Retrofit Intake:** Installation of continuous soffit venting (e.g., 3" strip or perforated soffit) to provide **>1,080 sq. in.** of Intake NFA to match the ridge vent.
2. **Balance the System:** Disconnect or block the SV-50s (or convert them to passive exhaust if intake is handled elsewhere).
3. **Verification:** A temporary "Blockage Test" (sealing the ridge vent with plastic for 2 weeks) can be performed to confirm that stopping the exhaust stops the ghosting accumulation.

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7. Validation

This diagnosis has been cross-referenced against building science principles regarding **Stack Effect**, **Bernoulli's Principle**, and **Manufacturer Specifications**.

References:

1. **Menzies SV-50 Specs ("Not for Intake"):** [Menzies Product Data]

2. **Lomanco Omni Ridge Specs (18 sq in/ft):** Lomanco Omni Data
3. **Ghosting & Air Movement:** ServiceMaster Restoration Guide
4. **IRC Ventilation Code (Balance Ratio):** Burgess Construction Consultants Analysis

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